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15.1 Least Squares as a Maximum Likelihood Estimator

Suppose that we are fitting N data points (x_i, y_i) , $i = 0, \dots, N-1$, to a model that has M adjustable parameters a_j , $j = 0, \dots, M-1$. The model predicts a functional relationship between the measured independent and dependent variables,

$$y(x) = y(x|a_0 \dots a_{M-1}) \quad (15.1.1)$$

where the notation indicates dependence on the parameters explicitly on the right-hand side, following the vertical bar.

What, exactly, do we want to minimize to get fitted values for the a_j 's? The first thing that comes to mind is the familiar least-squares fit,

$$\text{minimize over } a_0 \dots a_{M-1} : \sum_{i=0}^{N-1} [y_i - y(x_i|a_0 \dots a_{M-1})]^2 \quad (15.1.2)$$

But where does this come from? What general principles is it based on?

To answer these questions, let us start by asking, "Given a particular set of parameters, what is the probability that the observed data set should have occurred?" If the y_i 's take on continuous values, the probability will always be zero unless we add the phrase, "...plus or minus some small, fixed Δy on each data point." So let's always take this phrase as understood. If the probability of obtaining the data set is too small, then we can conclude that the parameters under consideration are "unlikely" to be right. Conversely, our intuition tells us that the data set should not be too improbable for the correct choice of parameters.

To be more quantitative, suppose that each data point y_i has a measurement error that is independently random and distributed as a normal (Gaussian) distribution around the "true" model $y(x)$. And suppose that the standard deviations σ of these normal distributions are the same for all points. Then the probability of the data set is the product of the probabilities of each point:

$$P(\text{data} | \text{model}) \propto \prod_{i=0}^{N-1} \left\{ \exp \left[-\frac{1}{2} \left(\frac{y_i - y(x_i)}{\sigma} \right)^2 \right] \Delta y \right\} \quad (15.1.3)$$

Notice that there is a factor Δy in each term in the product.

If we are Bayesians, we proceed by invoking Bayes' theorem, in the form

$$P(\text{model} \mid \text{data}) \propto P(\text{data} \mid \text{model}) P(\text{model}) \quad (15.1.4)$$

where $P(\text{model}) = P(a_0 \dots a_{M-1})$ is our prior probability distribution on all models. As often as not, we take a constant, *noninformative* prior. The most probable model, then, is the one that maximizes equation (15.1.3) or, equivalently, minimizes the negative of its logarithm,

$$\left[\sum_{i=0}^{N-1} \frac{[y_i - y(x_i)]^2}{2\sigma^2} \right] - N \log \Delta y \quad (15.1.5)$$

Since N , σ , and Δy are all constants, minimizing this equation is equivalent to minimizing (15.1.2).

If we are frequentists, we must get to the same destination by a more tortuous path (as is so often the case when Bayesian and frequentist methods coincide). We are not allowed to think about the notion of probability as applied to parameter sets, because, for frequentists, there is no statistical universe of models from which the parameters are drawn. We instead substitute a dictum: We identify the probability of the data given the parameters (which is computable as above), as the *likelihood* of the parameters given the data. This identification is entirely based on intuition. It has no formal mathematical basis in and of itself. Parameters derived in this way are called *maximum likelihood estimators*.

What we see is that least-squares fitting gives an answer that is both (i) the most probable parameter set in the Bayesian sense, assuming a flat prior, and (ii) the maximum likelihood estimate of the fitted parameters, in both cases *if* the measurement errors are independent and normally distributed with constant standard deviation. Notice that we made no assumption about the linearity or nonlinearity of the model $y(x|a_0 \dots)$ in its parameters $a_0 \dots a_{M-1}$. Just below, we will relax our assumption of constant standard deviations and obtain the very similar formulas for what is called “chi-square fitting” or “weighted least-squares fitting.” First, however, let us discuss further our very stringent assumption of a normal distribution.

For a hundred years or so, mathematical statisticians have been in love with the fact that the probability distribution of the sum of a very large number of very small random deviations almost always converges to a normal distribution. (For precise statements of this *central limit theorem*, consult [1] or other standard works on mathematical statistics.) This infatuation tended to focus interest away from the fact that, for real data, the normal distribution is often rather poorly realized, if it is realized at all. We are often taught, rather casually, that, on average, measurements will fall within $\pm\sigma$ of the true value 68% of the time, within $\pm2\sigma$ 95% of the time, and within $\pm3\sigma$ 99.7% of the time. Extending this, one would expect a measurement to be off by $\pm20\sigma$ only one time out of 2×10^{88} . We all know that “glitches” are much more likely than *that*!

In some instances, the deviations from a normal distribution are easy to understand and quantify. For example, in measurements obtained by counting events, the measurement errors are usually distributed as a Poisson distribution, whose cumulative probability function was already discussed in §6.2. When the number of counts going into one data point is large, the Poisson distribution converges toward

a Gaussian. However, the convergence is not uniform when measured in fractional accuracy. The more standard deviations out on the tail of the distribution, the larger the number of counts must be before a value close to the Gaussian is realized. The sign of the effect is always the same: The Gaussian predicts that “tail” events are much less likely than they actually (by Poisson) are. This causes such events, when they occur, to skew a least-squares fit much more than they ought.

Other times, the deviations from a normal distribution are not so easy to understand in detail. Experimental points are occasionally just *way off*. Perhaps the power flickered during a point’s measurement, or someone kicked the apparatus, or someone wrote down a wrong number. Points like this are called *outliers*. They can easily turn a least-squares fit on otherwise adequate data into nonsense. Their probability of occurrence in the assumed Gaussian model is so small that the maximum likelihood estimator is willing to distort the whole curve to try to bring them, mistakenly, into line.

The subject of *robust statistics* deals with cases where the normal or Gaussian model is a bad approximation, or cases where outliers are important. We will discuss robust methods briefly in §15.7. All the sections between this one and that one assume, one way or the other, a Gaussian model for the measurement errors in the data. It is quite important that you keep the limitations of that model in mind, even as you use the very useful methods that follow from assuming it.

Finally, note that our discussion of measurement errors has been limited to *statistical* errors, the kind that will average away if we only take enough data. Measurements are also susceptible to *systematic* errors that will not go away with any amount of averaging. For example, the calibration of a metal meter stick might depend on its temperature. If we take all our measurements at the same wrong temperature, then no amount of averaging or numerical processing will correct for this unrecognized systematic error.

15.1.1 Chi-Square Fitting

We considered the chi-square statistic once before, in §14.3. Here it arises in a slightly different context.

If each data point (x_i, y_i) has its own, known standard deviation σ_i , then equation (15.1.3) is modified only by putting a subscript i on the symbol σ . That subscript also propagates docilely into (15.1.5), so that the maximum likelihood estimate of the model parameters (and also the Bayesian most probable parameter set) is obtained by minimizing the quantity

$$\chi^2 \equiv \sum_{i=0}^{N-1} \left(\frac{y_i - y(x_i | a_0 \dots a_{M-1})}{\sigma_i} \right)^2 \quad (15.1.6)$$

called the “chi-square.”

To whatever extent the measurement errors actually *are* normally distributed, the quantity χ^2 is correspondingly a sum of N squares of normally distributed quantities, each normalized to unit variance. Once we have adjusted the $a_0 \dots a_{M-1}$ to minimize the value of χ^2 , the terms in the sum are not all statistically independent. For models that are linear in the a ’s, however, it turns out that the probability distribution for different values of χ^2 at its minimum can nevertheless be derived analytically, and is the *chi-square distribution for $N - M$ degrees of freedom*. We

learned how to compute this probability function using the incomplete gamma function in §6.2. In particular, equation (6.14.39) gives the probability Q that the chi-square should exceed a particular value χ^2 by chance, where $\nu = N - M$ is the *number of degrees of freedom*. The quantity Q , or its complement $P \equiv 1 - Q$, is frequently tabulated in appendices to statistics books, or it can be computed as $P = \text{Chisqdist}(\nu). \text{invcdf}(\chi^2)$ by the routine in §6.14.8. It is quite common, and usually not too wrong, to assume that the chi-square distribution holds even for models that are not strictly linear in the a 's.

This computed probability gives a quantitative measure for the goodness-of-fit of the model. If Q is a very small probability for some particular data set, then the apparent discrepancies are unlikely to be chance fluctuations. Much more probably either (i) the model is wrong — can be statistically rejected, or (ii) someone has lied to you about the size of the measurement errors σ_i — they are really larger than stated.

While above we were quick to poke fun at the frequentist's foundations for maximum likelihood estimation (or lack thereof), we must now take aim at strict Bayesians: There are no good fully Bayesian methods for assessing goodness-of-fit, that is, for comparing the probability of a best-fit model to that of a nonspecific alternative hypothesis like “the model is wrong.” The problem is that the strict application of Bayes theorem requires either (i) a comparison between two well-posed hypotheses (the *odds ratio*), or (ii) a normalization of the probability of the best-fit model against an integral of such probabilities over all possible models (the *normalizing constant*). In most situations neither of these is available. Sensible Bayesians usually fall back to p -value tail statistics like chi-square probability when they really need to know if a model is wrong.

Another important point is that the chi-square probability Q does not directly measure the credibility of the assumption that the measurement errors are normally distributed. It assumes they are. In most, but not all, cases, however, the effect of nonnormal errors is to create an abundance of outlier points. These decrease the probability Q , so that we can add another possible, though less definitive, conclusion to the above list: (iii) the measurement errors may not be normally distributed.

Possibility (iii) is fairly common, and also fairly benign. It is for this reason that reasonable experimenters are often rather tolerant of low probabilities Q . It is not uncommon to deem acceptable on equal terms any models with, say, $Q > 0.001$. This is not as sloppy as it sounds: Truly *wrong* models will often be rejected with vastly smaller values of Q , 10^{-18} , say. However, if day-in and day-out you find yourself accepting models with $Q \sim 10^{-3}$, you really should track down the cause.

If you happen to know the actual distribution law of your measurement errors, then you might wish to *Monte Carlo simulate* some data sets drawn from a particular model, cf. §7.3 – §7.4. You can then subject these synthetic data sets to your actual fitting procedure, so as to determine both the probability distribution of the χ^2 statistic and also the accuracy with which your model parameters are reproduced by the fit. We discuss this further in §15.6. The technique is very general, but it can also be slow.

At the opposite extreme, it sometimes happens that the probability Q is too large, too near to 1, literally too good to be true! Nonnormal measurement errors cannot in general produce this disease, since the normal distribution is about as “compact” as a distribution can be. Almost always, the cause of too good a chi-